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Thundercomm Turbox™ C410/C610 Open Kit Linux Enablement

Software Release Notes

Rev V1.2
July 06, 2021

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Revision History

Revision	Date	Description
1.0	Apr. 02, 2021	Initial release.
1.1	June 25, 2021	Update the following sections: • Chapter 3. Development Environment Setup • 4.2. Download LE SDK
1.2	July 06, 2021	Update the SDK code download command in 4.4.2.Download the LE SDK code.

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About This Document

- Illustrations in this documentation might look different from your product.
- Depending on the model, some optional accessories, features, and software programs might not be available on your device.
- Depending on the version of operating systems and programs, some user interface instructions might not be applicable to your device.
- Documentation content is subject to change without notice. Thundercomm makes constant improvements on the documentation of your computer, including this guidebook.
- Function declarations, function names, type declarations, attributes, and code samples appear in a different format, for example, `cp armcc armcpp`.
- Code variables appear in angle brackets, for example, `<number>`.
- Button, tool, and key names appear in bold font, for example, click **Save** or press **Enter**.
- Commands to be entered appear in a different font; on the host computer use $ as shell prompt, while on the target device use # as shell prompt, for example,

```
$ adb devices
# logcat
```

- Part of the code that does not contain instructions appear in a different format, for example,

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
```

- Folders and files are formatted in italic, for example, *turbox_flash_flat.sh*.

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Chapter 1. Introduction

This document provides information on detailed steps to obtain, build, and program QCS610 LE (Linux Enablement) SDK (Software Development Kit) into the Turbox™ C410/C610 open kit.

The Linux Enablement Software Development Kit supports the Linux operating system on the Thundercomm Turbox™ SOM. The LE SDK aims to provide a simple and flexible development platform to rapidly build IoT (Internet of Things) products with the QTI chipsets. The LE SDK provides the build framework, software components, tools, and sample applications to evaluate the platform and build software images for the target product.

The LE SDK is based on the Yocto Project 2.6 release and follows several of its development principles. The Yocto Project is used by the LE SDK to build framework.

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Chapter 2. System Requirements

Table 2-1 lists the hardware and software required to install and run the LA SDK.

Table 2-1. Hardware and software requirement to install and run the LE SDK

Hardware / Software Requirements	Source / Vendor	Function
Standalone hardware system minimum requirement: • 8 GB RAM • Quad core CPU ➲**NOTE:** System with lower hardware specifications will increase compilation time. About an hour is the compilation time for a system with 16 GB RAM and Intel i7-2600 @3.4 GHz.	\	SDK build machine
Ubuntu 16.04 Long Term Support (LTS) Linux distribution for 64 bit architecture	Ubuntu Community/ Canonical, Ltd.	SDK build host Operating System (OS)
Repository (Repo)	SDK Open Source Project	SDK source management tool
Python 2.7.12 with sqlite3 module	Python.org	Building subsystem

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Chapter 3. Development Environment Setup

This chapter introduces the setup of a Linux development environment on an Ubuntu host machine.

Prerequisites:

Ensure your building host meet the following requirements:

- Host machine Linux distro – 16.04 LTS
- Minimum 300 GB free disk space
- tar v1.27 or newer
- Privileges to install utilities
- Install the host packages

3.1. Ubuntu 16.04 (64-bit) installation

Step 1.Install Ubuntu 16.04 to the computer following the instructions at:
http://www.ubuntu.com/download/desktop/install-ubuntu-desktop.

Step 2.To update software, execute the following command:

```
$ sudo apt-get update
```

3.2. Tools and USB driver installation

This section provides instructions on installing **ADB** (Android Debug Bridge) (V1.0.41 or later), **Fastboot**, and USB drivers to your host computer.

The **Fastboot** is a tool used to flash images to the board by communicating with the boot loader.

The **ADB** is a tool installed on the host computer to access the device Unix shell, then help to push, modify, debug, and tweak device system files from the host computer.

3.2.1. ADB and Fastboot tools installation

To install **ADB** and **Fastboot**, execute the following commands.

```
$ sudo apt-get install android-tools-adb
$ sudo apt-get install android-tools-fastboot
```

3.2.2. USB driver installation and configuration

To install and configure USB drivers, follow the steps below:

Step 1.Set up the udev rules on a Linux computer.

1. Login as **root/sudo**.

   ```
   $ cd /
   ```

2. Navigate to the directory */etc/udev/rules.d/*.

   ```
   $ cd /etc/udev/rules.d/
   ```

3. Create the file *99-android.rules.*

   ```
   $ sudo touch 99-android.rules
   ```

4. Enter the following command get edit permission to file *99-android.rules*.

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```
$ sudo chmod 777 /etc/udev/rules.d/99-android.rules
```

5. Enter *99-android.rules* by running the following command.

```
$ vim 99-android.rules
```

6. Input `i` to enter edit mode and add the following content to the file.

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
SUBSYSTEM=="usb", ATTR{idVendor}=="05c6", MODE="0777", GROUP="adm"
```

7. Press **ESC** key to exit the edit mode.
8. Enter `:wq` to save the change and exit.

Step 2. Restart the udev service by executing the following commands.

```
$ sudo chmod a+r /etc/udev/rules.d/99-android.rules
$ sudo service udev restart
```

Step 3. Connect the board to your computer with an USB cable, and power on the target board to start the system. Wait for several seconds then input `$ adb devices`. The device is correctly detected with the following information displayed on your screen.

```
$ adb devices
List of devices attached
7050372c       device
```

➲ **NOTE:** Avoid connecting the device to the PC with a USB hub. Connect the device to the PC directly via a USB cable.

3.3. Required Building Environment

Follow the steps below to configure the building environment by install the proper version GCC/G++ and Linux package dependencies.

3.3.1. Java 8

Step 1. Check the version. Run the following commands to ensure Java 8 is installed.

```
$ java -version
openjdk version "1.8.0_131"
OpenJDK Runtime Environment (build 1.8.0_131-8u131-b11-0ubuntu1.16.04.2-b11)
OpenJDK 64-Bit Server VM (build 25.131-b11, mixed mode)
```

Step 2. Version update

With a 1.7 version preinstalled, run the following commands to upgrade it to version 1.8.

```
$ sudo apt-get update
$ sudo apt-get install openjdk-8-jdk
```

➲ **NOTE:** The Java 8 should be used along with Ubuntu 16.04.

3.3.2. Linux package dependencies

Step 1. Install the dependency libraries to the host computer.

```
$ sudo apt-get install git-core gnupg flex bison build-essential zip curl zlib1g-dev
gcc-multilib g++-multiliblibc6-dev-i386 lib32ncurses5-dev x11proto-core-dev libx11-
dev lib32z1-dev libgl1-mesa-devlibxml2-utilsxsltproc unzip fontconfig libxml-simple-
perl
```

Step 2. Set **bash** as default shell tool.

```
$ sudo rm /bin/sh
```

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```
$ sudo ln -sf /bin/bash /bin/sh
```

➲ **NOTE**: For more information on building environment dependencies, refer to AOSP documents at: *https://source.android.com/setup/build/initializing*.

3.3.3. Install repo

Run the following command to install repo.

```
$ sudo apt-get install repo
```

3.3.4. Configure git global variable

Configure user name and email.

```
$ git config --global user.email "<User Email>"
$ git config --global user.name "<User Name>"
```

➲ **NOTE:** Replace the `<User Email>` and `<User Name>` with the email address and user name you have registered at Thundercomm website.

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Chapter 4. Get Started with LE SDK

4.1. LE SDK information

Table 4-1. LE SDK Information

Baseline	Baseline Distribution	SDK Version
QCS610.LE.1.0	CS3 0.0.097.1	FC2.r001004

4.2. Block diagram

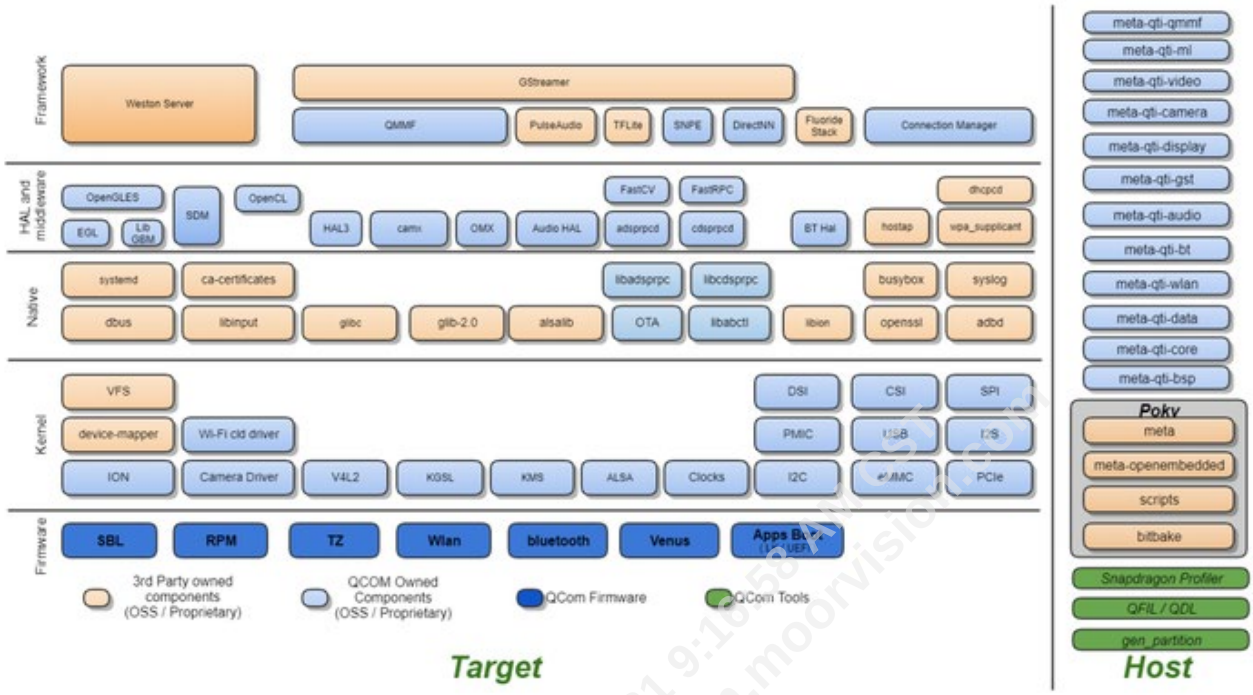


Figure 4-1. Block Diagram

4.3. SDK source tree structure

```
SDK.Turbox_C610.LE1.0_FC2/
├── about.html -> common/about.html          //SDK version info(Product,
Distribution,Components, …)
├── adsp_proc            //aDSP image is released as a binary and no build compilation is
needed
├── aop_proc             //AOP image is released as a binary and no build compilation is
needed
├── apps_proc
   ├── build-qti-distro-fullstack-virtualization-debug          /* Directory of temporary
                                                  fileand image files generated by
compilation */
   ├── disregard
   ├── downloads                        //Software packages required for compilation
   ├── poky                          //Yocto's recipe directory
   ├── prebuilt_HY11                    //Qcom privatized software package
   ├── setup-environment -> poky/qti-conf/set_bb_env.sh    //Script for set the build
environement
   ├── src                           /* AP source code, such as linux kernel, filesystem,
vendor,                                                                  OTA, etc. */
   ├── sstate-cache
   └── syncbuild.sh
```

```
├── boot_images           //BOOT image is released as a binary and no build compilation is
needed
├── BP-CODE                    //BP binary files
├── btfm_proc             //BTFM image is released as a binary and no build compilation is
needed
├── cdsp_proc             //cDSP image is released as a binary and no build compilation is
needed
├── common/config/emmc/partition.xml                          eMMC partition file
├── contents.xml -> common/contents.xml              //image contents file
├── modem_proc            //MODEM image is released as a binary and no build compilation
is needed
├── trustzone_images      //TZ image is released as a binary and no build compilation is
needed
├── turbox
│   └── tools
│       ├── build_script                        //SDK build script(android build, zip, …)
│       └── flash_tools                              //Flash tools files(qdl, …)
├── turbox_build.sh -> turbox/tools/build_script/common/turbox_build.sh       //SDK build
file.
├── turbox_flash.sh -> turbox/tools/build_script/device/turbox_flash.sh       //Fastboot
flash tool
├── venus_proc            //VIDEO image is released as a binary and no build compilation
is needed
├── wdsp_proc             //wDSP image is released as a binary and no build compilation is
needed.
└── wlan_proc             //WLAN image is released as a binary and no build compilation is
needed.
```

4.4. Download LE SDK

Follow the steps below to activate an account and download the SDK code.

4.4.1. Account activation

Add the account/password information to *$HOME/.netrc* by running the following command.

```
$ echo "machine partner.thundercomm.com login <User Email> password <password>" >> ~/.netrc
```

➲ **NOTE:** Please use the email and password you have registered at *www.thundercomm.com* to replace `<User Email>` and `<password>` in the above command.

4.4.2. Download the LE SDK code

Follow the steps below to download the LE SDK:

Step 1.Initialize the repository.

- In mainland China:

```
$ repo init -u https://partner.thundercomm.com/sdkgit/a/manifest -b
SDK.Turbox_C610.LE1.0 -m turbox-c610-le1.0-dev.release.FC2.r001004_cn.xml --repo-
url=https://partner.thundercomm.com/sdkgit/a/tools/repo --repo-branch=stable --no-
repo-verify
```

- Out of mainland China:

```
$ repo init -u https://partner.thundercomm.com/sdkgit/a/manifest -b
SDK.Turbox_C610.LE1.0 -m turbox-c610-le1.0-dev.release.FC2.r001004.xml --repo-
url=https://partner.thundercomm.com/sdkgit/a/tools/repo --repo-branch=stable --no-
repo-verify
```

Step 2.Synchronize the source code.

```
$ repo sync -c -j4 --no-tags
```

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4.5. SDK version update

If you have downloaded the old-version SDK, there are two ways to download the new SDK fast.

Option 1. Upgrade the local old folder to a new release

Navigate to the old local code folder. You can run the repo init command of new release as *4.2.2.Download the LE SDK code*, then run `repo sync -j4 -c --no-tags` to receive the new code.

Option 2. Download the new release to a new folder (recommended)

Create a new folder, and run `repo init` with `-reference` value.

For example:

```
$ repo init -u https://partner.thundercomm.com/sdkgit/a/manifest -b
SDK.Turbox_C610.LE1.0 -m turbox-c610-le1.0-dev.release.FC2.r001004.xml --repo-
url=https://partner.thundercomm.com/sdkgit/a/tools/repo --repo-branch=stable --
no-repo-verify --reference=<old_sdk_code_path>
$ repo sync -j4 -c --no-tags
```

➲ **NOTE:** In `--reference=<old_sdk_code_path>`, the `old_sdk_code_path` is the absolute path of your old SDK folder. For example:
`--reference=/home/host/turbox-c610-le1.0-dev.es.r000002`

4.6. Build image

To compile AP code, run the following script file according to your image version in the root directory.

- Build debug version:

```
$ ./turbox_build.sh -a -l -v debug
```

- Build perf version:

```
$ ./turbox_build.sh -a -l -v perf
```

➲ **NOTE**: UART debug cannot be used in perf/user version.

- Implement the following commands to rebuild kernel additionally (if necessary).

```
$ export SHELL=/bin/bash
$ cd apps_proc
$ MACHINE=qcs610-odk-64 DISTRO=qti-distro-fullstack-virtualization-debug
$ source poky/qti-conf/set_bb_env.sh
$ bitbake -fc virtual/kernel
$ bitbake -fc make_bootimg qti-multimedia-image
```

➲ **NOTE**: The default *boot.img* has verity enabled and cannot be flashed if only the kernel is rebuilt. Hence, when rebuilding only the kernel, one has to flash the *boot.img.noverity* which has verity disabled, refer to *turbox_flash.sh* to flash the *boot.img.noverity*.

- Implement the following commands to rebuild dtbo additionally (if necessary).

```
$ export SHELL=/bin/bash
$ cd apps_proc
$ MACHINE=qcs610-odk-64 DISTRO=qti-distro-fullstack-virtualization-debug
$ source poky/qti-conf/set_bb_env.sh
$ bitbake -fc cleanall linux-msm && bitbake linux-msm
```

4.7. Pack ROM package

Step 1.Run the following command to update common information according to your image version.

- **Debug version:**

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```
$ ./turbox_build.sh -u --uci-imf -v debug
```

- **Perf version:**

```
$ ./turbox_build.sh -u --uci-imf -v perf
```

➲ **NOTE:** The command is successfully executed with the following log displayed.

```
- update_common_info.py:======  UPDATE COMMON INFO COMPLETE=======
```

Step 2. Run the following command to pack the ROM package. You can find the packed ROM package at *turbox/output*. Replace `<debug/perf>` with debug or perf.

```
$ ./turbox_build.sh --zip_fastboot_build -l -v <debug/perf>
```

➲ **NOTE:** Customers can also get the ROM package provided by Thundercomm from:

- *FlatBuild_Turbox_C610_xx.xx_LE1.0.R.debug.FC2.r001004.zip*
- *FastbootBuild_Turbox_C610_xx.xx_LE1.0.R.debug.FC2.r001004.zip*
- *FlatBuild_Turbox_C610_xx.xx_LE1.0.R.perf.FC2.r001004.zip*
- *FastbootBuild_Turbox_C610_xx.xx_LE1.0.R.perf.FC2.r001004.zip*

4.8. Flash images

4.8.1. Flash Fastboot image

Step 1. Enter the Fastboot mode

Enter the following command when the device in ADB mode.

```
$ adb root
$ adb reboot bootloader
```

Power off the device, hold the volume down key. Press the power key about 5 seconds later, then release the **Volume Down** key.

➲ **NOTE**:

To check if the device has entered the Fastboot mode, run the following command:

```
$ fastboot devices
```

Successful with the following output displaying on the shell:

```
7050372c fastboot
```

Step 2. Flash image

Execute the commands below to flash images to the board.
Replace **`<folder name of image package>`** with the actual image package name of your product.

```
$ cd <folder name of image package>
$ bash ./turbox_flash.sh -r -v perf
```

➲ **NOTE**: Enter **`./turbox_flash -h`** to acquire detailed usage.

```
$ ./turbox_flash -h
Usage:
  bash ./turbox_flash.sh [options] [options]
  The images except gpt,userdata,persist will be flashed if no options input
Description:
  flash images with fasboot
Options:
  -h, --help   display this help message
  -v, --variant <debug/perf/user>    set the build variant.(default: debug)
```

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```
  -f, --force  gpt,persist and userdata will be flashed.
               User data and calibration data(BT/WIFI MAC address, etc..) will be
erased.
  -r, --reboot     reboot the device automatically after flashing completed.
  -b, --nonhlos    flash all bp images(xbl,rpm,ftfm,tz,dsp,modem,pmic).
  -a, --hlos       flash all ap
images(aboot,bootimage,recovery,system,vendor,cache,mdtp).
  --all       flash all images
Example:
  ./turbox_flash.sh
     Flash the images except gpt,userdata,persist.
  ./turbox_flash.sh --force
     Flash all images.
  ./turbox_flash.sh --hlos
     Flash all ap images.
```

4.8.2. Flash FlatBuild image

If the Open kit cannot boot up to enter the Fastboot mode, flash the FlatBuild image using *QDL*.

Step 1. Enter 9008 mode

Press and hold the **Force_USB_Boot** key, then plug in the USB device and power cable. When the 9008 USB device has been detected, release the **Force_USB_Boot** key.

To check if your device has entered the 9008 mode, run the following command:

```
$ lsusb
……
Bus 001 Device 042: ID 05c6:9008 Qualcomm, Inc. Gobi WireLAss Modem (QDL mode)
…….
```

Step 2. Start to flash the FlatBuild image

Check if the images are ready, then add the QDL tool to flash the images:

```
$ cd <folder name of image package>
$ chmod +x qdl
$ sudo ./qdl --storage emmc prog_firehose_ddr.elf  rawprogram*.xml patch*.xml
```

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Chapter 5. New Feature

Table 5-1. New Feature List

Feature	status
OTA	PASS
DTB Overlay	PASS
64-bit Userspace	PASS
Azure and Docker	PASS
DM-Verity/ SELinux/ KASan/ FDE/ VBLE/ Non-Root Processes	PASS
Introducing GStreamer Framework	PASS
RTSP Streaming	PASS
Staggered HDR v2	PASS
Defog/ IRLED/ LDC/ EIS	PASS
Dual Independent Cameras	PASS
OpenGL ES 3.2	PASS
OpenCL 2.1	PASS
AI Inferencing using TFlite 2.0	PASS
AI Inferencing using SNPE	PASS

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Chapter 6. Known Issues

- The system cannot enter sleep mode.
- **Face Detection** demo is not stable.
- SNPE does not work on DSP.
- HDMI IN does not work on C410 SOM.

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